Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 6: Approved Solutions

Biconditionals, and a Textbook Claim Worth Checking

Covers Chartrand & Zhang, Section 1.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything before reading any solution. Problem 8 is unusual — read its instructions carefully.

Part I: How to Practice This Material

A biconditional is two claims wearing one coat

The only two facts you need

1. $P \Leftrightarrow Q$ is true exactly when P and Q have the **same truth value**. To evaluate one, find both truth values and compare. Nothing else is required.
2. $P \Leftrightarrow Q$ is $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. To *prove* one, you must prove **both** implications. To *refute* one, a single failing direction is enough.

Three mistakes to expect

1. **Thinking a true biconditional needs both sides true.** Two false statements also agree, so the biconditional holds.
2. **Proving one direction and stopping.** Half a proof of an "iff" is not a proof.
3. **Assuming a printed claim is correct.** Problem 8 exists because of this one.

Part II: Practice Problems

Section A — Evaluating biconditionals

1. Determine whether each is true or false, with a one-line reason.

(a) 11 is prime if and only if 12 is composite.

(b) $2^4 = 16$ if and only if $4^2 = 8$.

(c) $\sqrt{2}$ is rational if and only if 0 is negative.

(d) 6 is even if and only if 6 is prime.

Approved solution

(a) True. Both sides are true, so they agree.
(b) False. $2^4 = 16$ is true; $4^2 = 16 \neq 8$, so the right side is false. Different truth values.
(c) True. Both sides are false — $\sqrt{2}$ is irrational and 0 is not negative — so they agree.
(d) False. 6 is even (true) but not prime (false).

Part (c) is the one to look at. Nothing connects $\sqrt{2}$ to the sign of zero, and both halves are false, yet the biconditional is true. As with the implication, the definition consults only truth values.

2. For an integer n, let $P(n)$: n is divisible by 6, and $Q(n)$: n is divisible by both 2 and 3.

(a) Evaluate $P(n) \Leftrightarrow Q(n)$ for $n = 12$, $n = 9$, and $n = 5$.

(b) Do you believe $P(n) \Leftrightarrow Q(n)$ holds for every integer n? Say what you would have to prove to establish it.

Approved solution

(a)

n	$P(n)$	$Q(n)$		$P(n) \Leftrightarrow Q(n)$
12	T	T		**T**
9	F	F	**T**	(divisible by 3, not by 2)
5	F	F		**T**

(b) It does hold for every integer. To establish it you must prove *two* implications: if n is divisible by 6 then it is divisible by 2 and by 3; and if n is divisible by 2 and by 3 then it is divisible by 6. Three confirming values are not a proof — the same lesson as Practice Set 1. What they do give is confidence that looking for a proof is worthwhile.

Section B — Saying it in words

3. For an integer n, let $P(n)$: $5n + 3$ is even, and $Q(n)$: n is odd. State $P(n) \Leftrightarrow Q(n)$ (a) using "if and only if" and (b) using "necessary and sufficient."

Approved solution

(a) $5n + 3$ is even if and only if n is odd.
(b) $5n + 3$ being even is necessary and sufficient for n to be odd.

Aside, worth checking: the claim is in fact true. If n is odd then $5n$ is odd, so $5n + 3$ is even; if n is even then $5n$ is even and $5n + 3$ is odd.

4. Rewrite each as a biconditional using "if and only if."

(a) A valid session token is necessary and sufficient for access to be granted.

(b) If the queue is empty the worker sleeps, and if the worker sleeps the queue is empty.

Approved solution

(a) Access is granted if and only if the session token is valid.
(b) The worker sleeps if and only if the queue is empty.
Part (b) is the definition applied directly: an implication together with its converse is a biconditional. Any time you see both directions asserted, they can be collapsed into one "iff."

Section C — Connections back to Chapter 1

5. Use a truth table to verify that $P \Leftrightarrow Q \equiv (P \wedge Q) \vee \big((\sim P) \wedge (\sim Q)\big)$.

P	Q	$P \Leftrightarrow Q$	$P \wedge Q$	$(\sim P) \wedge (\sim Q)$	$(P \wedge Q) \vee \big((\sim P) \wedge (\sim Q)\big)$
T	T	T	T	F	T
T	F	F	F	F	F
F	T	F	F	F	F
F	F	T	F	T	T

Approved solution

The two columns agree in all four rows, so the equivalence holds.
Read the right-hand side aloud: *either both are true, or both are false.* That is what "same truth value" means, spelled out with the connectives from Week 1 — which is a reminder that $\Leftrightarrow$ adds no new power, only convenience.

6. Explain, in two or three sentences, why $\sim(P \Leftrightarrow Q) \equiv P \oplus Q$ without building a new truth table.

Approved solution

$P \Leftrightarrow Q$ is true exactly when P and Q have the same truth value, so its negation is true exactly when they have *different* truth values. But "exactly one of the two is true" is precisely the definition of $P \oplus Q$. Hence the two are equivalent.
No table is needed because both connectives were defined by the same question — do the two sides agree? — answered in opposite directions.

Section D — Writing

7. In five to eight complete sentences, explain why proving a biconditional requires two separate arguments, while proving an implication requires only one. Illustrate with the claim "an integer n is odd if and only if n^2 is odd," and say specifically what the two arguments would have to establish.

Approved solution

Model answer. By Definition 1.52, $P \Leftrightarrow Q$ is the conjunction of $P \Rightarrow Q$ and $Q \Rightarrow P$, and a conjunction is true only when both parts are true. So establishing a biconditional means establishing both implications; proving one leaves the conjunction unsettled. An implication, by contrast, is a single claim with a single thing to show. For the claim about odd integers, the first argument must show that if n is odd then n^2 is odd. The second must show the reverse: if n^2 is odd then n is odd. These are genuinely different tasks — the second is usually handled by proving its contrapositive, that if n is even then n^2 is even. Stopping after the first would leave open the possibility of an even integer with an odd square, which is exactly what the second argument rules out.

Full credit requires naming both directions *specifically* for the odd-integer claim, not just saying "you prove it both ways." The strongest answers notice that the second direction is naturally attacked by contrapositive.

Section E — Challenge (optional, and unusual)

8. Exercise 24(e) in Section 1.3 of your textbook instructs you to *verify* the following by truth table:

$$(P \Rightarrow Q) \wedge (Q \Rightarrow R) \;\equiv\; P \Rightarrow R.$$

Build the table. Report exactly what you find, and do not adjust your answer to match what the book expects.

Approved solution

The claim is false as printed. The two sides differ in two rows.

P	Q	R	$P \Rightarrow Q$	$Q \Rightarrow R$	$(P \Rightarrow Q) \wedge (Q \Rightarrow R)$	$P \Rightarrow R$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	**F**	**T**
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	F	T	F	**F**	**T**
F	F	T	T	T	T	T
F	F	F	T	T	T	T

They disagree at TFT and FTF. So the stated equivalence does not hold. Parts (a) through (d) of that exercise are all correct; only (e) is misstated.

What is true is the one-directional version:

$$\Big((P \Rightarrow Q) \wedge (Q \Rightarrow R)\Big) \Rightarrow (P \Rightarrow R)$$

is true in all eight rows. In words: if P implies Q and Q implies R, then P implies R. That is the chaining rule you would expect, and it is what the exercise was reaching for. Chaining lets you conclude the right-hand side *from* the left; it does not make the two interchangeable.

Why this problem is on the page. Every practice set this semester has told you that a claim must be checked rather than assumed. That applies to claims in print, including this textbook and including anything I hand you. The table is the authority.

Instructor notes

- **Problem 8 uses a genuine error in the textbook.** Section 1.3 Exercise 24(e) is misstated; (a)–(d) are fine. I verified all five parts. **Do not assign 24(e) as ordinary homework** — students who trust the book will "verify" it by fudging a row. Used deliberately, as here, it is the best possible closing lesson for a unit built on checking claims. Decide whether you want that; if not, drop Problem 8 and the set still stands.
- **Problem 1(c)** is the biconditional analogue of the "$2+2=5$" implication: two unrelated false statements produce a true biconditional. Expect resistance.
- **Problem 7** previews Chapter 3 directly. Students who notice that the reverse direction wants a contrapositive are ready for proof techniques.
- **Problem 6** should be answered in prose, not with a table. If most students build one anyway, that signals they are still computing rather than reasoning from definitions.
- Reserved for Homework 3: Section 1.3 Exercises 2, 4, 6, 8, 14 and Section 1.4 Exercises 2, 4, 6, 8 — all even, so no printed answers. Odd-numbered exercises have answers in the back (pp. 596–597) and are used only for self-check.